Exact Stationary Equilibria in Discounted Perfect-Information Games

A three-player arithmetic construction and FIXP-completeness

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Abstract

For every fixed number of players at least three, exact stationary equilibrium computation in rational discounted perfect-information stochastic games is FIXP-complete under separable-linear reductions. Hardness already holds at discount 1/2, with at most two actions per state. We give explicit signed-affine, clipping, and reciprocal modules, including their behavior at every equilibrium in a graph with feedback. Unconditional interval registers and a uniform reward bound prevent circular assumptions about the inputs. A three-state reciprocal module changes the player whose value carries a scalar; its output is exact even while one other value coordinate is only approximately calibrated. Root states give the source coordinates as one minus a single continuation probability, including at the endpoints. A bounded circuit for three-player strategic Nash supplies the hardness source. For membership, we construct a fixed-point circuit whose value-evaluation divisions are defined on the whole domain and whose fixed points satisfy every-state Bellman optimality against arbitrary behavioral deviations.

1 Problem, result, and exact reductions

A perfect-information stochastic game has finite states S, a controller $c(s) \in \{1, \dots, n\}$ at each state, a finite nonempty action set $A(s)$, rational rewards $r_i(s, a)$, rational transition probabilities $P(s' \mid s, a)$, and a rational discount $0 < \gamma < 1$. All tables are explicitly listed in binary. A stationary profile assigns $\sigma(\cdot \mid s) \in \Delta(A(s))$. Its normalized value is

$$V_i^\sigma(s) = (1-\gamma)\mathbb{E}_{s,\sigma} \sum_{t=0}^{\infty} \gamma^t r_i(s_t, a_t). \tag{1.1}$$

The required equilibrium is optimal for every player from every initial state against every unilateral behavioral, possibly history-dependent, deviation. This is the C73-82 relation. Exact real outputs are understood in the fixed-point search model, not as necessarily rational strings or as approximations to a residual.

We use the standard definition of FIXP by fixed points of rational algebraic circuits with operations $+, -, \times, \div, \max, \min$, defining continuous self-maps of compact rational-polytope domains, with every operation defined on the entire domain. Our maps use cubes. A separable-linear reduction has a polynomial-time instance map and decodes each source coordinate as $a_i y_{j(i)} + b_i$, with rational coefficients, from one target coordinate. These are the SL reductions of Etessami and Yannakakis [1]. In particular, a nonlinear Bellman solve is not permitted as our hardness decoder.

Theorem 1 (Exact classification)**.** *For each fixed integer $n \geq 3$, computing a stationary every-state equilibrium of a rational discounted perfect-information stochastic game is* FIXP*-complete under* SL *reductions. The hardness restriction may simultaneously require $\gamma = 1/2$ and at most two actions at every state. The same classification holds when the evaluated value vector is included in the output.*

The proof covers arbitrary recurrent transition structures. It is not restricted to acyclic state graphs, rational equilibria, approximate incentives, or equilibrium from a selected state. Hansen and Nie [2] classify the two-player case and leave the three-or-more-player exact classification open in their Section 6. Their three-player cyclic construction motivates the value-carrying states used here. The reduction below supplies signed arithmetic, channel conversion, simultaneous feedback control, and an exact coordinate decoder. The earlier catalogue attempt [3] treats triangular transition structures and pure-policy verification; it does not provide this classification.

2 Bellman semantics and the global guard

For a profile σ, let r_i^σ and P^σ be its expected reward vector and transition matrix. They satisfy

$$V_i^\sigma = (1-\gamma) r_i^\sigma + \gamma P^\sigma V_i^\sigma. \tag{2.1}$$

For any vector V_i, write

$$Q_i(s, a; V_i) = (1-\gamma) r_i(s, a) + \gamma \sum_{s'} P(s' \mid s, a) V_i(s'). \tag{2.2}$$

Lemma 2 (Every-state verification)**.** *A stationary profile is an equilibrium in the stated behavioral sense if and only if its evaluated values satisfy*

$$Q_{c(s)}(s, a; V_{c(s)}^\sigma) \leq V_{c(s)}^\sigma(s) \qquad (s \in S, a \in A(s)).$$

At a controlled state, every action used with positive probability then has value equal to the maximum action value.

Proof. An equilibrium admits the deviation that changes the first action and then resumes the profile, giving necessity. Evaluation is the profile-weighted mean of the action values, so every supported action must attain the maximum. Conversely, fix a player and an arbitrary behavioral deviation. At its controlled states the displayed inequality applies conditionally on the full past; at other states evaluation gives equality. Iterating conditional expectations for T steps bounds the deviation's normalized truncated reward plus $\gamma^T V_i^\sigma(s_T)$ by the starting value. The values are bounded and $\gamma^T \to 0$. Letting $T \to \infty$ proves the required inequality, from every state. □

All hardness modules have $\gamma = 1/2$. A port has a type (a, o, b), one of $(1, 2, 3), (2, 3, 1), (3, 1, 2)$: actor, scalar owner, and passive coordinate. A restored port for u has value vector $(1/2, u, 1)$ in that order. During the proof we will establish scalar and actor identities without assuming that all passive coordinates are already exactly one.

Each type has an absorbing sink Z_{aob} with per-stage rewards $(0, 0, 1)$. There is also an all-zero absorbing sink. Every sink has one action. An absorbing state with rewards $(1/2, c, 1)$ is a constant port for any rational c.

The finite construction first specifies its rational intervals, affine coefficients, scalar quit rewards, and reward shifts. Each later reward has a bound independent of the small transition parameters:

in a term pA, for example, use $|pA| \le |A|$ for $p \le 1$. Choose a rational $M_0 \ge 4$ dominating all these bounds and constant rewards, and put

$$\delta = \frac{1}{16(M_0+1)}, \qquad \kappa = \frac{\delta}{4-\delta}. \tag{2.3}$$

All resulting stage rewards have absolute value at most M_0, and therefore so do every player's normalized values under *every* profile. No correct-circuit assumption is involved. Each monitor's transition mass to nonsink states will be at most δ.

2.1 The two-state linear template

A module has an output state v controlled by a, a monitor w controlled by o, and same-type input ports. At v, action Q has rewards $(1, 2K, 1)$ and moves to Z_{aob}; action C has rewards $(0, 0, 1)$ and moves to w. Thus the value of Q is $(1/2, K, 1)$. Denote the probability of C by x.

At w, the actor reward is 4 on action L and 0 on action R. The passive reward is 1 on both. The scalar rewards and transitions will be specified below. Every nonsink transition goes to v or a same-type input port, and their total mass on either action is at most δ. The remaining mass goes to Z_{aob}.

Lemma 3 (Uniform guards and passive attenuation)**.** *At every equilibrium, pure L at w forces pure C at v, and pure R forces pure Q. No exact input actor coordinate is required. For any strategies, if $e_s = V_b(s) - 1$, then*

$$|e_v| \le \kappa \max_{\text{input } s} |e_s| \le \kappa(M_0+1) < \frac{1}{4}, \tag{2.4}$$

with zero on the right when there are no input ports.

Proof. The global value bound gives

$$Q_a(w, L) \ge 2 - \delta M_0/2 > 1, \qquad Q_a(w, R) \le \delta M_0/2 < 1.$$

At v, continuing discounts the monitor value by $1/2$, whereas quitting gives $1/2$. This proves the guards. For the passive coordinate, quitting at v gives one, continuing has error $e_w/2$, and each monitor action has error one half its transition-weighted feedback and input errors. This holds for every monitor mixture. Consequently

$$|e_v| \le \frac{\delta}{4}|e_v| + \frac{\delta}{4} \max_{\text{input } s} |e_s|.$$

Rearrangement proves the first inequality; the second uses the global value bound. Formula (2.3) gives the final strict bound. □

No later step requires the monitor to mix with probability $1/2$. The guards and passive estimate apply to arbitrary input values and arbitrary feedback paths.

3 Exact affine gates and unconditional registers

Lemma 4 (Strict signed-affine gate)**.** *Suppose same-type input scalar values satisfy $u_i \in [l_i, h_i]$, and let $F = A + \sum_i c_i u_i$, with rational data. There is a rational two-state module that, at every equilibrium, has scalar value F, actor value $1/2$, and the passive estimate (2.4). It permits signed coefficients and amplification.*

Proof. Compute the full Cartesian-box bounds L_F, U_F of this affine form. They do not rely on correlations between correctly evaluated circuit inputs. Set $K = \max(1, U_F + 1)$. Before choosing any transition parameter, take a positive rational D such that

$$D > 4(|L_F| + 1) + |A| + \sum_{c_i > 0} c_i \max(|l_i|, |h_i|). \tag{3.1}$$

Take $p = \delta/(1 + \sum_i |c_i|)$. On L, the scalar reward is $-D$, with transition p to v and $p(-c_i)$ to input i for each $c_i < 0$. On R, the scalar reward is $-D + pA$, with transition pc_i to input i for $c_i > 0$. Fill the sink probabilities and use the other rewards of the template. Both transition masses are at most δ.

Write $z = V_o(v)$, and let R_0 be the scalar action value of R at w. Then

$$Q_o(w, L) - Q_o(w, R) = \frac{p}{2}(z - F), \tag{3.2}$$

$$R_0 = \frac{1}{2}\left(-D + pA + p\sum_{c_i > 0} c_i u_i\right), \qquad \frac{R_0}{2} < L_F \le F. \tag{3.3}$$

The last inequality follows from (3.1) and $p \le 1$.

If $x = 0$, then $z = K > F$, so the monitor strictly chooses L and the guard forces C, a contradiction. If $x = 1$, write λ for the monitor's L probability. Evaluation gives

$$(1 - p\lambda/4)(z - F) = R_0/2 - F < 0.$$

Thus the monitor strictly chooses R, forcing Q, another contradiction. Hence $0 < x < 1$, so the actor value is exactly $1/2$. A pure monitor would again force an endpoint. The monitor therefore mixes, and (3.2) implies $z = F$. Lemma 3 supplies the passive estimate. Every equation concerns values in the actual global equilibrium, even if input paths return to this module. □

Lemma 5 (Unconditional clipping)**.** *Let $F = A + \sum_i c_i u_i$ with $c_i \ge 0$. Without any input scalar range assumption, a two-state module has value* $\mathrm{clip}(F, 0, 1) = \min(1, \max(0, F))$ *at every equilibrium. Its passive coordinate satisfies* (2.4)*.*

Proof. Use $K = 1$. Give L scalar reward zero and transition p to v. Give R scalar reward pA and transitions pc_i to the inputs. Choose $p = \delta/(1 + \sum_i c_i)$, and retain the actor and passive rewards of the template. The scalar action values are $pz/2$ and $pF/2$. The monitor's Bellman optimality yields

$$z = (1 - x) + \frac{xp}{4}\max(z, F). \tag{3.4}$$

If $0 < x < 1$, the guards force the monitor to mix, so $z = F$ and $z = (1 - x)/(1 - px/4) \in (0, 1)$. If $x = 0$, then $z = 1$, and $F < 1$ would force L and then C; hence $F \ge 1$. If $x = 1$, strict R would force Q, so $z \ge F$. Equation (3.4) becomes $z = pz/4$, giving $z = 0$ and $F \le 0$. These cases include the equality endpoints and prove the claim. The passive estimate does not depend on optimality. □

Definition 6 (Interval register)**.** For a nondegenerate rational interval $[l, h]$, first apply Lemma 5 to $(u - l)/(h - l)$, then apply Lemma 4 to $l + (h - l)t$, with $t \in [0, 1]$. The resulting port represents $\mathrm{clip}(u, l, h)$, belongs to $[l, h]$, has actor value $1/2$, and satisfies (2.4). For $l = h$, use a constant port.

Every logical wire in the construction receives an unconditional register at its declared intended interval. This is essential: the lower endpoint of a raw clip need not have actor value $1/2$, and the registered scalar range must hold before the functional circuit has been shown correct. A raw affine gate is calibrated to the independent Cartesian input box and is only then registered to its intended logical range. For example, a correlated expression known eventually to be a square cannot use nonnegativity as its raw affine bound.

4 A reciprocal that changes value channels

Lemma 7 (Robust exact reciprocal conversion)**.** *Let a registered type* $(1,2,3)$ *port have scalar* $u \in [m, M]$*, where* $0 < m \le M$ *are rational. A rational constant-size construction using the same three players produces a type* $(3,1,2)$ *port for* $1/u$*. At every equilibrium, its actor and passive coordinates are exactly* $1/2$ *and* 1*.*

Proof. A strict affine preprocessor gives scalar $w = 4u/m \in [4, 4M/m]$. Its player 1 value is exactly $1/2$, while its player 3 value B satisfies $|B-1| \le \kappa(M_0+1) < 1/4$. Thus the input need not already have an exactly calibrated passive coordinate.

Create three states s_1, s_2, s_3, controlled by players $1, 2, 3$. Continuing has zero rewards and moves to the next state cyclically. Quitting at s_1, s_2 goes to the all-zero sink, with respective reward vectors

$$(1, 1/2, 4), \qquad (4, 1, 1/2).$$

Quitting at s_3 has reward vector $(-1/2, 1/4, 0)$ and moves to the preprocessor port. The effective quit values are consequently

$$q_1 = (1/2, 1/4, 2), \quad q_2 = (2, 1/2, 1/4), \quad q_3 = (0, H, t), \tag{4.1}$$

where

$$H = 1/8 + w/2 \ge 17/8, \qquad t = B/2 \in [3/8, 5/8]. \tag{4.2}$$

Let x_j be the continuation probability at s_j.

We first exclude all boundary strategies; this step does not presume any port identities beyond the preceding bounds. If $x_1 = 0$, player 3 must continue, receiving $1 > t$; then player 2 must quit, since continuing gives $1/16 < 1/2$; then player 1 must continue, receiving $1 > 1/2$, a contradiction. If $x_2 = 0$, player 1 must continue; player 3 must then quit since its continuation gives $1/16 < t$; and player 2 must continue since $H/2 > 1/2$. If $x_3 = 0$, player 2 must continue, player 1 must quit because its continuation gives zero, and player 3 must continue. These also contradict the assumed pure quit.

If $x_1 = 1$, player 3's continuation action has value

$$(1 - x_2)/16 + (x_2/8)V_3(s_3).$$

Were continuing optimal there, its value would be at most $1/16 < t$, contrary to the available quit. Hence $x_3 = 0$, already excluded. If $x_2 = 1$, player 1's continuation value is $x_3 V_1(s_1)/8$, with zero intervening exit value, so quitting is strict and $x_1 = 0$. If $x_3 = 1$, the analogous continuation calculation for player 2 bounds its value by $1/16 < 1/2$, forcing $x_2 = 0$. Thus

$$0 < x_j < 1 \qquad (j = 1, 2, 3) \tag{4.3}$$

at every equilibrium, including one in a larger recurrent game.

The fixed quits of actors 1 and 2 now give

$$V_1(s_1) = 1/2, \qquad V_2(s_2) = 1/2, \qquad V_2(s_3) = 1.$$

The last identity is player 2's indifference at s_2. Evaluation at s_1 gives $V_2(s_1) = 1/4$, regardless of x_1. Writing $z = V_1(s_3)$, evaluation at s_3 then yields

$$z = x_3/4, \qquad 1 = H(1 - x_3) + x_3/8, \qquad z = \frac{1}{4} - \frac{7}{32(H - 1/8)} = \frac{1}{4} - \frac{7}{16w}. \tag{4.4}$$

These identities do not involve t. In particular, exact reciprocal arithmetic does not require first proving that every passive coordinate in the whole graph equals one. Also $z \in [9/64, 1/4)$, and the passive player 2 value at the output state s_3 is exactly one.

Register z in $[0, 1/4]$, apply the strict signed-affine map

$$\frac{64}{7m}\left(\frac{1}{4} - z\right) = \frac{1}{u}, \tag{4.5}$$

and register to $[1/M, 1/m]$. The proved bounds make these registrations inactive on their intended scalar values. They restore actor 3 to $1/2$. Their passive coordinate remains exactly one: their upstream passive value at s_3 is one, and the linear evaluation equations in Lemma 3 preserve that constant. This proves the claim. □

Cyclic relabeling gives conversion from each scalar owner to the preceding one. Three conversions return to the original scalar channel and perform one reciprocal; two perform an identity while changing channel, and four give the other identity-channel change. All intermediate positive intervals are explicitly reciprocated and registered. This uses a constant number of states per reciprocal and exactly the same three players.

Example 8 (A rational check of the converter)**.** For $m = u = 1$ and exact input fields, $w = 4$, $H = 17/8$, and $t = 1/2$. The continuation probabilities are

$$x_1 = 8/15, \qquad x_2 = 128/247, \qquad x_3 = 9/16.$$

The state value vectors are

$$(1/2, 1/4, 1), \quad (1, 1/2, 1/4), \quad (9/64, 1, 1/2).$$

Substitution verifies both action values for each controller and all evaluation equations. The postprocessor gives $(64/7)(1/4 - 9/64) = 1$. This example checks the constants.

5 Roots with an affine coordinate decoder

Lemma 9 (Root equation, including endpoints)**.** *Let F be the scalar of an unconditional $[0, 1]$ register. There is a root state v, with continuation probability x, such that every equilibrium satisfies*

$$V_2(v) = F = 1 - x. \tag{5.1}$$

This conclusion does not assume correctness of the circuit that supplies the raw input to the final register.

Proof. Use type $(1, 2, 3)$. A strict affine module produces $-F \in [-1, 0]$. At v, use the linear template with $K = 1$. At its player 2 monitor, R has scalar reward zero and goes to the type sink. Action L has scalar reward zero and transitions p to v and p to the $-F$ port, where $p = \delta/2 > 0$. Retain actor rewards $4, 0$ and passive reward one. If $z = V_2(v)$, Bellman optimality gives

$$z = (1 - x) + \frac{xp}{4}\max(z - F, 0). \tag{5.2}$$

Its right-hand side is nonnegative, so $z \geq 0$. If $z > 1$, then $z > F$ and this equation gives

$$z = \frac{1 - x - xpF/4}{1 - xp/4} \leq \frac{1 - x}{1 - xp/4} \leq 1,$$

a contradiction. If $z > F$, the monitor chooses L strictly, forcing $x = 1$, and $(1 - p/4)z = -pF/4 \leq 0$, again a contradiction. If $z < F$, the monitor chooses R strictly, forcing $x = 0$ and $z = 1 \geq F$, also impossible. Thus $z = F$, and (5.2) gives $z = 1 - x$. □

The functional circuit reads a registered $[0, 1]$ copy of the root. Lemma 9 makes this registration inactive. Its scalar lies in the cube before circuit correctness is proved. The raw root's actor coordinate need not equal $1/2$ at an endpoint; the register restores it. The final decoder uses the strategy coordinate $1 - x$ directly.

6 The simultaneous compiler

A bounded arithmetic circuit here means a rational straight-line circuit on a coordinate cube, with a valid rational intended interval on every logical wire. Its operations are signed affine maps, clipping, and reciprocals whose intended intervals have positive lower endpoints. Coefficient, interval, and fan-in encodings have polynomial length. Section 7 supplies such circuits from a known complete problem; we do not assume magnitude bounds for arbitrary untyped algebraic circuits.

Proposition 10 (Compilation with feedback)**.** *Given a bounded arithmetic map $F : [0,1]^d \to [0,1]^d$ of this form, one can construct in polynomial time a rational three-player perfect-information game, with discount $1/2$ and at most two actions per state, so that every stationary every-state equilibrium decodes to a fixed point of F by $z_i = 1 - x_i$, where x_i is one root's continuation probability.*

Proof. Create one root template per coordinate. Its feedback port is linked after building the straight-line functional circuit. That circuit reads registered root scalars in the common scalar-owner 2 channel.

Replace a logical affine gate by Lemma 4, calibrated to the full independent input box, then register its output at its intended logical interval. Replace a reciprocal by three successive Lemma 7 conversions, including their preprocessors, postprocessors, and interval registers. This returns the scalar to owner 2. Implement logical clipping with Lemma 5 and rational affine normalization, restoring the output with a register. Constant intervals use constant ports. Every logical wire is thus unconditionally registered. Give each final output an unconditional $[0,1]$ register, form its strict negative, and link the root monitor of Lemma 9.

Choose all endpoints, scalar shifts, and rewards before the small probabilities. A strict affine module may, for example, use any rational D one unit greater than the right side of (3.1). Its scalar rewards are bounded by $D+|A|$ and $2|K|$, independent of $p \le 1$. Clip rewards are bounded by $|A|$ and two. Converter stage rewards are fixed rational constants; their possibly large input scalars are represented by bounded-input affine modules, not inserted as reward data. Take M_0 to dominate these finitely many bounds, four, and all constant-port rewards. Now choose δ from (2.3), the affine probabilities $p = \delta/(1 + \sum |c_i|)$, and root probabilities $p = \delta/2$. All sink probabilities are nonnegative and each transition row sums to one. There is no inverse power of p or δ in a reward, so this order of construction is noncircular.

Consider any global stationary every-state equilibrium. Its reward bound first gives the uniform actor guards and passive attenuation. All unconditional registers then satisfy their scalar range guarantees without correct-circuit assumptions. Consequently every strict affine gate has its exact scalar and actor coordinates simultaneously; in particular this applies to every converter's positive preprocessor. Those preprocessors have valid positive scalar ranges, exact actor coordinate $1/2$, and passive coordinates within $1/4$ of one. Lemma 7 therefore applies to *all* converters. It forces full mixing and exact reciprocal scalar identities without inductively assuming that any upstream converter is already correct. If an affine consumer reads a raw converter value, the prescribed unconditional register precedes that consumer. The converter's proved range subsequently shows that register is inactive.

Independently, the final output registers and negative-output gates allow Lemma 9 to prove $z_i = F_i^{\mathrm{actual}} = 1 - x_i$ and $z \in [0,1]^d$. Now perform ordinary topological induction through the *functional straight-line circuit*, treating these root scalars as its fixed input. Its inputs have the declared ranges; every raw arithmetic identity is valid; and each intended wire registration is inactive on the correct value. This proves $F_i^{\mathrm{actual}} = F_i(z)$ and hence $F(z) = z$.

The stochastic game itself was never assumed acyclic. All local arguments used current equilibrium evaluation and Bellman optimality, which already include deviations at all of a player's

controlled states. An input continuation is not an exogenous payoff that is held unchanged under an otherwise profitable full deviation.

For size, each affine or register module has constant many states and a transition list linear in its fan-in. A same-channel reciprocal uses constant many such modules and three triads. Cartesian affine bounds, reward shifts, M_0, and the small probabilities use polynomially many rational operations on the supplied polynomial-bit data. No step requires root isolation, numerical equilibrium, or an accuracy search. Thus the game has polynomial size and rational bit length. Its equilibrium set is nonempty, for example by the fixed-point membership proof in Section 8. The decoder is separable affine by construction. □

7 A bounded complete source

We reduce exact three-player strategic-form Nash, which is FIXP-complete under SL reductions by Theorem 18 of Etessami and Yannakakis [1]. The input gives explicit rational payoff tables. For each player separately, let l be its minimum payoff and $D = \max(1, \max r - l)$; replace every entry by $(r - l)/D$. This is one positive affine transformation of the player's *whole* payoff function, and preserves best replies. A constant whole-player table becomes zero. Individual action rows are never normalized separately. All transformed rewards lie in $[0, 1]$.

7.1 Retraction from the cube to simplexes

For $x \in [0,1]^m$, let $x_{(1)} \geq \cdots \geq x_{(m)}$ be the sorted coordinates and put

$$\theta = \max_{1 \leq k \leq m} \frac{\sum_{j=1}^{k} x_{(j)} - 1}{k}, \qquad \Pi_i(x) = \max(x_i - \theta, 0). \tag{7.1}$$

Lemma 11 (Polynomial retraction)**.** *Formula* (7.1) *defines a continuous piecewise-linear retraction onto Δ_m, with a polynomial-size rational max/min/affine circuit. Its useful intermediate ranges have polynomial-bit endpoints.*

Proof. Let k attain the maximum. Its average formula implies $x_{(k)} \geq \theta$: otherwise removing that last entry would increase the average; for $k = 1$ this is immediate. If $k < m$, maximality after appending the next entry gives $x_{(k+1)} \leq \theta$. Thus exactly the positive terms among the first k contribute to $\sum_i \max(x_i - \theta, 0)$, and their sum is one. This proves that $\Pi(x) \in \Delta_m$. If x is already in the simplex, every prefix sum is at most one and the full prefix equals one, so $\theta = 0$ and $\Pi(x) = x$.

A fixed sequence of adjacent compare-exchange operations sorts m numbers using at most a quadratic number of comparators; each comparator is implemented by $\min, \max$. Prefix sums lie in $[0, m]$, the displayed threshold lies in $[-1, 1]$, and the final coordinates lie in $[0, 1]$. Rational division by the integer k is a fixed affine coefficient. Continuity and polynomial circuit size follow. □

Apply this retraction blockwise to the three players. If $\sigma = \Pi(x)$, define

$$g_{i,a}(\sigma) = \max\{0, U_i(a, \sigma_{-i}) - U_i(\sigma)\}, \tag{7.2}$$

$$N_{i,a}(\sigma) = \frac{\sigma_{i,a} + g_{i,a}(\sigma)}{1 + \sum_b g_{i,b}(\sigma)}, \qquad F(x) = N(\Pi(x)). \tag{7.3}$$

This is a cube map with image in the product of simplexes.

Lemma 12 (Fixed points and bounds)**.** *Every fixed point of F is a strategic Nash equilibrium. The map has a polynomial-size bounded arithmetic circuit of the kind used in Proposition 10.*

Proof. A fixed point belongs to the product of simplexes, so its retraction is itself. In a player's block put $G = \sum_a g_{i,a}$. The fixed point equations imply $G\sigma_{i,a} = g_{i,a}$. If $G > 0$, every action in the support has strictly positive original payoff advantage. But their σ_i-weighted advantage sum is $\sum_a \sigma_{i,a} U_i(a, \sigma_{-i}) - U_i(\sigma) = 0$, a contradiction. Hence all gains vanish, which is exactly Nash optimality.

Expected payoffs are finite sums of products of at most three strategy coordinates and listed rational payoffs. Their circuits have polynomial size in the explicit table size. Action and profile payoffs lie in $[0, 1]$; gains lie in $[0, 1]$; a numerator in (7.3) lies in $[0, 2]$, and a denominator lies in $[1, m_i + 1]$. Partial sums can use their larger independent-box bounds, and be registered at their valid logical ranges. We now give explicit operations sufficient to compile all these circuits.

For $t \in [0, 2]$, let $q = 1 + t \in [1, 3]$. Then

$$d = \frac{1}{q} - \frac{1}{q+1} \in [1/12, 1/2], \qquad t^2 = 12\frac{1}{12d} - 3q + 1. \tag{7.4}$$

Indeed $1/d = q(q + 1)$, and the last expression is $q(q + 1) - 3q + 1 = (q - 1)^2$. Its reciprocal inputs lie in $[1, 3]$, $[2, 4]$, and $[1, 6]$. For $u, v \in [0, 1]$,

$$uv = \frac{(u+v)^2 - u^2 - v^2}{2}.$$

Thus products use only a constant number of affine and positive reciprocal operations. Other bounded products are rationally rescaled first. For bounded arguments, $\max(a, b) = b + \mathrm{clip}(a - b, 0, C)$, where C is a known upper bound on their positive difference; min follows by subtraction. If $C = 0$, use the constant zero for the clipped term.

Every reciprocal interval has a strictly positive rational lower endpoint. All intervals have polynomial-bit endpoints. Each raw affine module still uses the *Cartesian* range: for example, the raw expression $12r - 3q + 1$, with $r \in [1/6, 1]$ and $q \in [1, 3]$, is bounded on that box by $[-6, 10]$, before registration to the logically valid square range $[0, 4]$. Likewise a sum whose intended value is at most one may need a raw box bound two. These larger bounds remain polynomial, and the induction establishing intended ranges uses only identities on the source cube. This gives the required bounded circuit. □

Hardness part of Theorem 1. Apply Proposition 10 to (7.3) using Lemma 12. Every returned target equilibrium has decoded coordinates $z_i = 1 - x_i$ forming a fixed point of F, and hence a source Nash equilibrium. These are precisely allowed SL output maps. The construction uses three players, discount $1/2$, and at most two actions at a state.

For a fixed $n > 3$, give each added player zero rewards everywhere, and add one isolated one-action absorbing state controlled by that player, with zero rewards for all players. Old states never reach the new ones. The old incentive constraints and decoder are unchanged, and the new states impose only trivial every-state conditions. This proves hardness for every fixed $n \geq 3$. □

8 Whole-domain membership

We give membership as well, so the theorem is a classification rather than only a hardness statement. Let an arbitrary input game have reward bound $B_0 = \max(1, \max_{i,s,a} |r_i(s, a)|)$. Project cube coordinates onto each controlled-state action simplex, using Lemma 11, obtaining σ. Its values solve

$$(I - \gamma P^\sigma)V_i = (1 - \gamma)r_i^\sigma. \tag{8.1}$$

Lemma 13 (No undefined division)**.** *Gaussian elimination without row pivoting in* (8.1) *has strictly positive pivots for every cube input after projection. It therefore gives a polynomial-size rational algebraic circuit defined on the whole domain.*

Proof. Initially the matrix has nonpositive off-diagonal entries and row sums exactly $1-\gamma > 0$. In particular each diagonal is positive. Suppose these properties hold at an elimination stage, with pivot $a_{11} > 0$. Its Schur complement has off-diagonal entries $a_{ij} - a_{i1}a_{1j}/a_{11} \le 0$. If r_i denotes the current row sum, the new row sum for row i is

$$r_i - \frac{a_{i1}}{a_{11}} r_1 \ge r_i \ge 1 - \gamma,$$

since $a_{i1} \le 0$. Thus the properties persist and every subsequent diagonal pivot is at least $1-\gamma$. Forward elimination and back substitution use a polynomial number of additions, multiplications, and divisions by these positive pivots. They compute the unique solution; equivalently uniqueness follows from the convergent Neumann series $\sum_{k\ge 0} \gamma^k (P^\sigma)^k$. □

At each controlled state define the evaluated action advantages

$$d_s(a) = Q_{c(s)}(s, a; V^\sigma_{c(s)}) - V^\sigma_{c(s)}(s)$$

and update its mixed action by

$$T_s(a) = \frac{\sigma(a \mid s) + \max(d_s(a), 0)}{1 + \sum_b \max(d_s(b), 0)}. \tag{8.2}$$

This maps the product of action simplexes into itself. Composing its input with the cube projection gives a continuous rational algebraic cube map, with every division defined everywhere. Brouwer's theorem gives a fixed point. At a fixed point the action-weighted advantage sum at each state is zero by (2.1); the same argument as in Lemma 12 therefore excludes every positive advantage. Lemma 2 gives precisely the required every-state equilibrium against arbitrary behavioral deviations. Conversely any such equilibrium is fixed by (8.2).

The construction has polynomial circuit size in the explicit game encoding, including the number of players when it is not fixed. Projection and evaluation are defined on the entire compact cube. For an output relation including the values, append coordinates that ignore their input values and output $(V_i^\sigma(s) + B_0)/(2B_0)$. Equation (1.1) bounds these in $[0, 1]$. At a fixed point they are the evaluated values, which decode by coordinatewise affine maps. The strategy coordinates are already coordinates of the fixed point. This proves FIXP membership and completes Theorem 1.

9 Encoding and provenance

All module data are rational. The magnitude bounds and intervals in the three-player strategic Nash reduction are polynomially bounded in its explicit table dimensions, with coefficients of polynomial bit length. Conservative Cartesian affine bounds and the choice (3.1) have polynomial bit length as well. Thus M_0, δ, and every transition probability have polynomial bit length. Each module has at most two actions per controlled state; multiple outgoing transition destinations do not create extra actions. Exactly three player identities are reused in every channel conversion. The reduction is a finite mathematical algorithm and does not call an equilibrium solver as a subroutine.

If bounded nonnegative stage rewards are desired, apply the common positive affine transformation $r \mapsto (r + M_0)/(2M_0)$ after the hardness construction. Every player's normalized total payoff undergoes the same positive affine transformation under every strategy, so all equilibrium strategies and the decoder are unchanged. The gate proof is evaluated before this optional normalization.

The completeness source and definition of SL reductions are taken from the primary full manuscript of Etessami and Yannakakis, whose Theorem 18 supplies the exact strategic Nash

classification. The stochastic-game baseline and the previously open multi-player classification are checked against Hansen and Nie, version 1.

This manuscript does not claim an executed general compiler or a numerical solution of a large constructed game.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

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